VIRAJ THAKKAR

Ph.D. Candidate, Arizona State University

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My research spans Storage Systems, Disaggregated Infrastructure, and AI for Systems; from teaching LLMs to tune, to keeping disaggregated data secure and caches consistent. Feel free to reach out if something catches your eye :)

RESEARCH INTERESTS

Data Infrastructure: LSM-based Key-Value Stores (RocksDB, LevelDB); Disaggregated Compute, Memory, and Storage; Storage for Quantum Computing; Data Confidentiality.

Intelligent Storage: AI/ML for Storage Systems; Storage Systems for AI/ML; Real-Time Storage Systems for High-Velocity and High-Volume Data.

EDUCATION

Ph.D. Computer Science — Aug. 2023 - Ongoing
Arizona State University — *Advisor: Dr. Zhichao Cao*
ASU Fulton Fellowship Award Recipient — 3.95 / 4.00

M.S. Computer Science — Jan. 2022 – Aug. 2023
Arizona State University — *Advisor: Dr. Zhichao Cao*
Thesis: Optimizing Consistency and Performance Trade-off in Distributed LSM-KVS. [PDF] — 3.87 / 4.00

B.Tech. Information Technology — Jan. 2017 – Sept. 2021
K. J. Somaiya College of Engineering, India — 8.56 / 10.00

HONORS, GRANTS, AND AWARDS

ASU Nominee for Google PhD Fellowship Award	*2026*
ACM SIGMOD Travel Grant	*2025, 2026*
ASU SCAI, Graduate College, and GSG Travel Grant	*2025, 2026*
OpenAI Researcher Access Program Credit Award	*2025, 2026*
HotStorage 2024 SK Hynix Best Paper Award	*2024*
USENIX FAST Travel Grant	*2024, 2025, 2026*
ASU Fulton Fellowship	*2023 - 2027*

PEER-REVIEWED PUBLICATIONS

13. [**ProTools '26**] **Viraj Thakkar**, Timo Sokler, John Bryne, MJ Son, Sekwon Lee, Harumi Kuno, Adita Dhakal. "Probe Less, Learn More: Characterizing and Correcting eBPF Instrumentation Overhead." 8th Workshop on Programming and Performance Visualization Tools. *Co-located with Supercomputing 2026*. *To Appear.*
12. [**SYSTOR '26**] **Viraj Thakkar**, Qi Lin, Kenanya Prasetyo, Raden Wisjnunandono, Jin Wang, Achmad Kistijantoro, Reza Rachmadi, Zhichao Cao. "TellyTune: Exploring the Limits of LLM-Driven Tuning for LSM-Based Key-Value Stores." *19th ACM International Systems and Storage Conference. [PDF]*
11. [**SeQureDB '26**] **Viraj Thakkar**, Dongha Kim, Hokeun Kim, Zhichao Cao. "ANCHOR: A Vision for Secure Persistent Key-Value Stores in Disaggregated Data Centers." *1st SeQureDB Workshop on Secure & Private Data Management, 2026. Research Track Workshop Paper. Co-located with ACM SIGMOD 2026. To Appear.*
10. [**SIGMOD '26**] Qi Lin, Gangqi Huang, Te Guo, Chang Guo, **Viraj Thakkar**, Zhichen Zhu, Jianguo Wang, Zhichao Cao. "O3-LSM: Maximizing Disaggregated LSM Write Performance via Three-Layer Offloading." *Proceedings of ACM Conference on Management of Data (SIGMOD), Research Track Full Paper, 2026. [PDF]*
9. [**HPE TechCon '26**] John Bryne, **Viraj Thakkar**, Matthias Frank, Elias Klett, Harumi Kuno, Sekwon Lee, Clarete Crasta, Abhishek Dwaraki, David Emberson, Ramya Rao, Amitha C, Chinmay Ghosh, Prono Shome, Shreyas Basri. "CacheFS." *Published at HPE internal conference. 82 of 1036 (8%) Acceptance Rate. Close Access.*

8. [**SIGMOD '25**] **Viraj Thakkar**, Dongha Kim, Yingchun Lai, Hoekun Kim, Zhichao Cao. "SHIELD: Encrypting Persistent Data of LSM-KVS from Monolithic to Disaggregated Storage." *Proceedings of ACM Conference on Management of Data (SIGMOD), Research Track Full Paper, 2025*. *[PDF]*
7. [**HotStorage '24**] **Viraj Thakkar**, Madhumitha Sukumar, Jiaxin Dai, Kaushiki Singh, and Zhichao Cao. 2024. "Can Modern LLMs Tune and Configure LSM-based Key-Value Stores?" *In Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems, July 08, 2024*. *ACM, Santa Clara CA USA, 116–123*. *[PDF]*
BEST PAPER AWARD!
6. [**SIGMOD '24**] Qiaolin Yu, Chang Guo, Jay Zhuang, **Viraj Thakkar**, Jianguo Wang, Zhichao Cao. "CaaS-LSM: Compaction-as-a-Service for LSM-based Key-Value Stores in Storage Disaggregated Infrastructure." *Proceedings of ACM Conference on Management of Data (SIGMOD), Research Track Full Paper, 2024. [PDF]*
5. [**RPA '23**] Ashwini Dalvi, **Viraj Thakkar**, Smit Moradiya, Aditya Vedpathak, Irfan Siddavatam, Faruk Kazi, S. G. Bhirud. "*FLASH: Web-Form's Logical Analysis & Session Handling Automatic Form Classification and Filling on Surface and Dark Web*." Robotic Process Automation (2023): 61-100. *[PDF]*
4. [**HRAI-HCI '22**] Irfan Siddavatam, Ashwini Dalvi, Abhishek Patel, Aditya Panchal, Aditya Vedpathak, **Viraj Thakkar**. "*ML-Enabled Informed Intervention for Crowdsourcing-Based Optimization of Medical Resources.*" Handbook of Research on Applied Intelligence for Health and Clinical Informatics. IGI Global, 2022. *[PDF]*
3. [**ICACIT '21**] Ashwini Dalvi, Irfan Siddavatam, **Viraj Thakkar**, Aditya Vedpathak, Abhishek Patel. "*Recognizing Child Unsafe Apps Through User Reviews on the Google Play Store.*" Advanced Computing and Intelligent Technologies (pp. 111-120). Springer, Singapore 2022. *[PDF]*
2. [**ICAST '21**] Irfan Siddavatam, Ashwini Dalvi, **Viraj Thakkar**, Aditya Vedpathak, Smit Moradiya, Apoorva Jain. "*Explainability using decision trees and Monte Carlo simulations.*" Proceedings of the 4th International Conference on Advances in Science & Technology (ICAST2021). *[PDF]*
1. [**IBSSC '21**] Ashwini Dalvi, Irfan Siddavatam, **Viraj Thakkar**, Apoorva Jain, Faruk Kazi, Sunil Bhirud. "*Link Harvesting on the Dark Web*." IEEE Bombay Section Signature Conference (IBSSC), pp. 1-5. IEEE, 2021. *[PDF]*

PEER-REVIEWED POSTERS AND ABSTRACTS

5. [**HPTS '26**] **Viraj Thakkar**. "Broad Reasoning but Narrow Control for an End-to-End KVS Tuner." *21st International Workshop on High Performance Transaction Systems (HPTS).* 2026.
4. [**FAST '26**] Zhenjie Sun, **Viraj Thakkar**, Avani Wildani, Zhichao Cao. "Beyond ARC – Introducing LLM-driven Adaptive Policy Replacement." *24th USENIX Conference on File and Storage Technologies, 2026.*
3. [**CUG '26**] John Bryne, Myungjun Son, Harumi Kuno, Aditya Dhakal, Sekwon Lee, **Viraj Thakkar**, Clarete Crasta, David Emberson. "Characterizing sustained low-level message latency." *Conference for the Cray User Group, 2026.*
2. [**FAST '25**] Yibo Zhao, **Viraj Thakkar**, Zhichao Cao, Zaoxing Liu. "NetLSM: Enabling an In-Network Approach for Scheduling LSM-KVS Operations." *23rd USENIX Conference on File and Storage Technologies, 2025.*
1. [**FAST '24**] Madhumitha Sukumar, Jiaxin Dai, Kaushiki Singh, Vikriti Lokegaonkar, **Viraj Thakkar**, Zhichao Cao. "LLM-assisted Automatic-Configuration and Tuning Framework for LSM-based Key-Value Stores." *22nd USENIX Conference on File and Storage Technologies, 2024.*

PREPRINTS / UNDER REVIEW

2. [**arXiv '25**] **Viraj Thakkar**, Qi Lin, Kenanya Keandra Adriel Prasetyo, Raden Haryosatyo Wisjnunandono, Achmad Imam Kistijantoro, Reza Fuad Rachmadi, Zhichao Cao. "ELMo-Tune-V2: LLM-Assisted Full-Cycle Auto-Tuning to Optimize LSM-Based Key-Value Stores." *arXiv:2502.17606.* 2025. *[PDF]*
1. [**arXiv '25**] Qi Lin, Zhenyu Zhang, **Viraj Thakkar**, Zhenjie Sun, Mai Zheng, Zhichao Cao. "StorageXTuner: An LLM Agent-Driven Automatic Tuning Framework for Heterogeneous Storage Systems." *arXiv:2510.25017.* 2025. *[PDF]*

INVITED TALKS

9. "Breaking Up The KVS: Fast, Secure, Tunable Storage under Disaggregation." *HPE Labs Invited Talk.* October 2026.
8. "How to Tune your Dragon." *Amazon Invited Talk.* August 2026.

7. "ANCHOR: A Vision for Secure Persistent Key-Value Stores in Disaggregated Data Centers." *1st SeQureDB Workshop on Secure & Private Data Management, 2026.*
6. "O3-LSM: Maximizing Disaggregated LSM Write Performance via Three-Layer Offloading." *ACM Conference on Management of Data (SIGMOD), 2026.*
5. "SHIELD: Encrypting Persistent Data of LSM-KVS from Monolithic to Disaggregated Storage." *ACM Conference on Management of Data (SIGMOD), 2025.*
4. "NetLSM: Enabling an In-Network Approach for Scheduling LSM-KVS Operations." *23rd USENIX Conference on File and Storage Technologies, 2025.*
3. "Can Modern LLMs Tune and Configure LSM-based Key-Value Stores?" *16th ACM Workshop on Hot Topics in Storage and File Systems HotStorage'24, CA, 2024.*
2. "LLM-assisted Automatic-Configuration and Tuning Framework for LSM-based Key-Value Stores." *22nd USENIX Conference on File and Storage Technologies, 2024.*
1. "Explainability using decision trees and Monte Carlo simulations." *4th International Conference on Advances in Science & Technology, 2021.*

PATENTS AND INVENTION DISCLOSURES

3. Harumi Kuno, **Viraj Thakkar**, John Byrne, Sekwon Lee, "Shared-Daemon Offload Path for FUSE-Established Files Backed by Remote Memory." *Under HPE Internal Disclosure Review, Patent filing upon Approval.*
2. John Bryne, **Viraj Thakkar**, Harumi Kuno, Sekwon Lee. "A Global, Strongly Consistent, Client-Sided Caching Mechanism." *Under HPE Internal Disclosure Review, Patent filing upon Approval.*
1. John Bryne, **Viraj Thakkar**, Harumi Kuno, Sekwon Lee. "Federated Coherence Management for a Globally Shared Cache." *Under HPE Internal Disclosure Review, Patent filing upon Approval.*

ACADEMIC SERVICES

Program Committee,

- *TechCon 2027 (Internal Conference at HPE)*

Shadow Program Committee,

- *VLDB 2027*
- *SIGMETRICS 2026*
- *EuroSys 2025*

Artifact Evaluation Committee,

- *FAST 2026*
- *FAST 2025*
- *SIGMOD 2025*

MENTORSHIP

2026	*HPE Labs Dual-Study Intern*	Timo Sokler *(Characterizing and Understanding eBPF Overheads)*
	MS Student	Anushwar Sharma *(The Fusion of LLMs and ML for LSM-KVS tuning)*
	MS Student	Asmi Pulgam *(A Study of Linux File System Evolution)*
2025	*HPE Labs Dual-Study Intern*	Elias Klett *(Benchmarking FUSE as an In-Memory Filesystem)*
	MS Student	Sampada Nemade *(Modeling Traces using Statistical Methods)*
2024	*MS Thesis Student*	Harry Deccoco *(Optimizing Post-deletion performance in LSM-KVS)*
2023	*Undergrad Project Group*	Madhumita Sukumar; Kaushiki Singh *(LLM-driven tuning for LSM-KVS)*
	MS Project Group	Jiaxin Dai *(LLM-driven tuning for LSM-KVS)*
2020	*Undergrad Robotics Club*	Anish Pawar; Kamal Rohra; Dhruv Joshi *(Simulating object physics in MATLAB)*

EXPERIENCE

HPE Labs, Systems Architecture Lab *California, USA*

System Software Research Associate Intern Aug. 2025 - Present

- Explored FUSE, Linux's Filesystem in Userspace for potential slowdowns due to kernel-scheduling and identified potential optimizations that are currently under development.
- Developed a prototype RDMA-based in-memory filesystem on top of FUSE to experiment on different solutions.

System Software Research Associate Intern May 2025 – Aug. 2025

- Performed a background survey about existing distributed caching/storage services like Redis and Memcached.
- Collaborated with the team to design and develop a federated token management solution that maintains cache coherence for a global distributed client-side cache.

Amazon *Washington D.C., USA*

Software Development Engineer Intern May 2023 – Aug. 2023

- Established product documentation to seamlessly onboard future team members.
- Collaborated with the team to develop a full-stack application (AWS Lambda, DynamoDB, S3 buckets, ECS, SQS CloudFormation) that provides an interface to check for copyright infringements for products on Amazon.

Somaiya Vidyavihar, IN & Children's Investment Fund Foundation (CIFF), U.K. *Mumbai, IN*

Junior Researcher Sept. 2019 – Mar. 2020

- Led team of six to develop an automated tool identifying Legal Cases about Online Child Sexual Abuse (OCSA) and Exploitation (OCSE) in India, presenting a comprehensive work to leading voices across the country.
- Collaborated with the Indian National Commission for Protection of Child Rights on a report that was submitted to the Rajya Sabha (Upper House) of the Indian Parliament.

PROJECTS IN PROGRESS

Automating LLM-Assisted Tuning for LSM-KVS

- LSM-KVS (like RocksDB) have 200+ interrelated options that need deep understanding to be tuned correctly.
- Researched and developed a framework that automates predicting workloads and setting up tuning parameters for LSM-KVS by leveraging corpus ingestion and inferencing abilities of Modern LLMs.

Leveraging Programmable Network Switch for pre-emptive LSM-KVS Operations

- LSM-KVS exposes options that can be dynamically modified at runtime to improve performance.
- Explored and developed a controller to enhance LSM-KVS performance using programmable network switches (P4 Switches) to perform operations and online configuration modifications for LSM-KVS.

Integrating Encryption into LSM-KVS for Monolith and Disaggregated Infrastructure

- Encryption of persistent data in LSM-KVS provides data confidentiality and is essential as deployments move from monolith to multi-tenant disaggregated infrastructures.
- Designed and developed scalable and decentralized encryption design for LSM-KVS that works from monolith to disaggregated infrastructure while embedding practices like encryption key rotation into LSM-KVS components.

SOCIAL CONTRIBUTIONS AND ACHIEVEMENTS

CGroup-Manager, A Python library to start and monitor a Linux process using cgroups. *[URL]* *2024*

Stack Overflow, *[URL]* *2023*

- 1200+ Reputation with 50+ answers. *Top 15% contributor in 2023*.

Hackathon by SVB & ASU *2022*

- Developed a full stack web application that connects Investors and Startups. *Runner-Up Prize*.

Memories, A Python library for Image Segmentation. *[URL]* *2022*

Internship and Hackathon Organizer, Mumbai, IN *2020 - 2021*

- Designed and organized a college-wide online internship for 100+ interns over 2 semesters.
- Worked with the college's computer science society to manage an annual College Hackathon.

Qubit by Qubit, a course on Quantum Computing by IBM Quantum and The Coding School *2021*

CoView, Interactive COVID-19 Portal for KEM Hospital, Mumbai, IN *2020*

- Implemented interactive dashboard for KEM Hospital to monitor patients in the COVID-19 pandemic.
- CoView was deployed and used for over 7 months during the pandemic. Featured in local news *[URL]*.

Team KJSCE Robocon, Mumbai, IN *2018 - 2020*

- Spearheaded a team of 40+ individuals and mentored members to take over the team in coming years.
- Achieved National Rank 8 and Winner of Best Solution Idea amongst 50+ teams in "DD-Robocon 2020."